M2 CAUSAL INFERENCE · ALBERT SCHOOL

Does a Fed rate hike move the S&P 500?

An observational causal study of FOMC decisions, 1994–2026.

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THE QUESTION

When the Fed raises rates at a meeting, does the S&P 500 move differently than it would have if the Fed had held instead?

We compare meetings where the Fed hiked to meetings where it held — using only information known the day before.

The setup

UNIT

One FOMC decision

Feb 1994 – Oct 2026

TREATMENT

Hike vs. hold

52 hikes · 179 holds

OUTCOME

S&P 500 return

2 and 21 days after

$$ATT = E[\,Y(1) - Y(0) \mid T = 1\,]$$

Who it's for: meetings where the Fed **actually hiked** — not a hypothetical where it hiked every time (that would be the ATE).

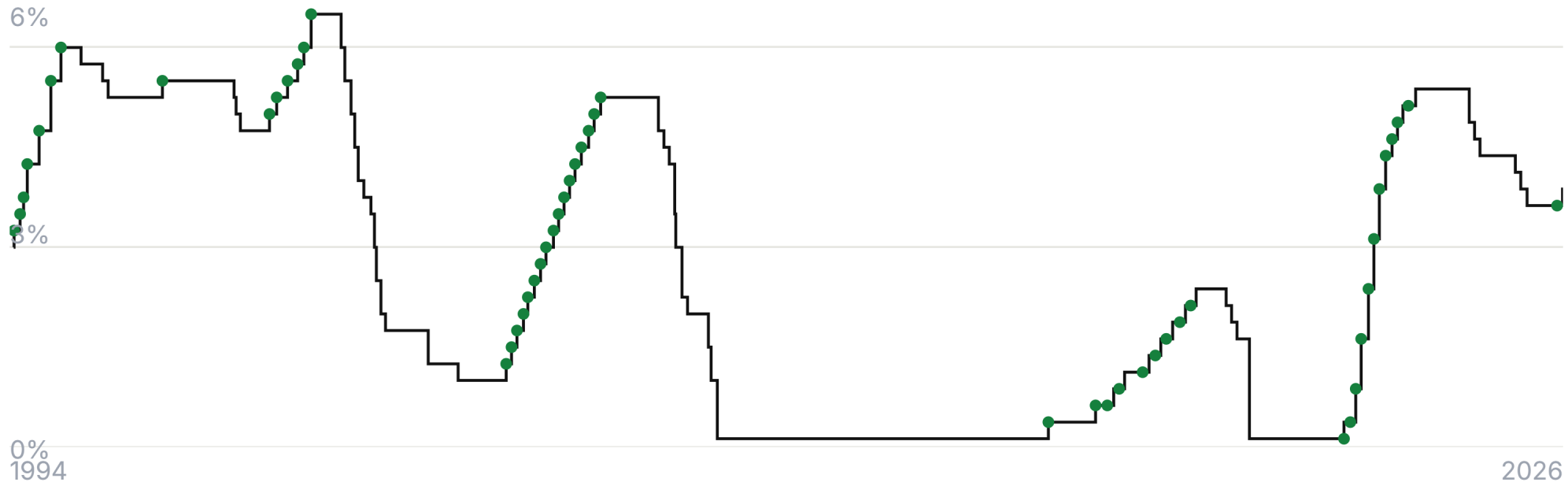
WHY IT MATTERS

A treasury with stock-market exposure must decide whether to hedge or wait as a hike approaches.

It can't change the Fed's decision — only its own reaction. It needs exactly this number: what a real hike does, compared with no hike.

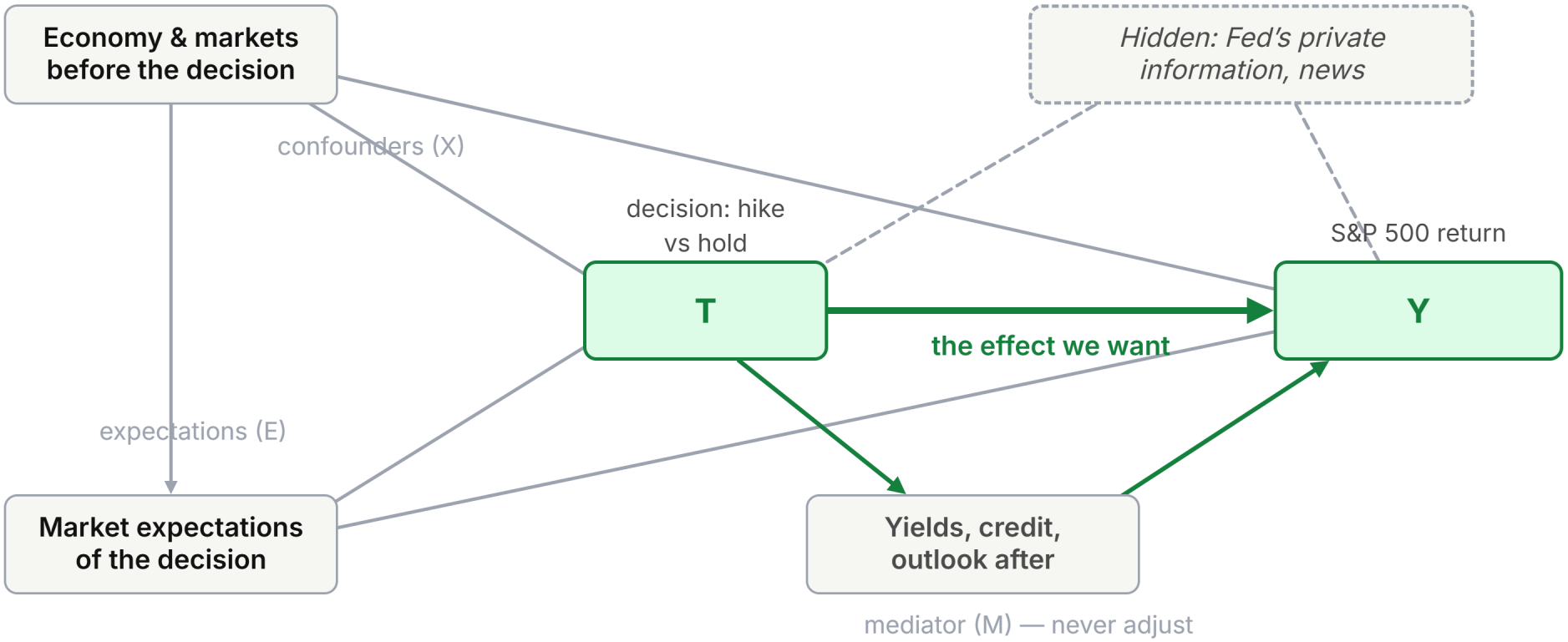
Our data

The Fed's target rate, every decision, 1994–2026



● each green dot is a hike — 52 in total, against 179 holds and 40 cuts

The causal graph



Choosing the method

Was treatment randomly assigned?	No
Several treated units, before & after? (DiD)	No
A clean, untouched donor mix? (Synthetic control)	No
Are the main confounders observed in the data?	**Yes**

→ Observational study

The key assumption

Once we account for what was known the day before, a hike and a hold are **as good as randomly assigned**.

Market expectations
yields vs. target

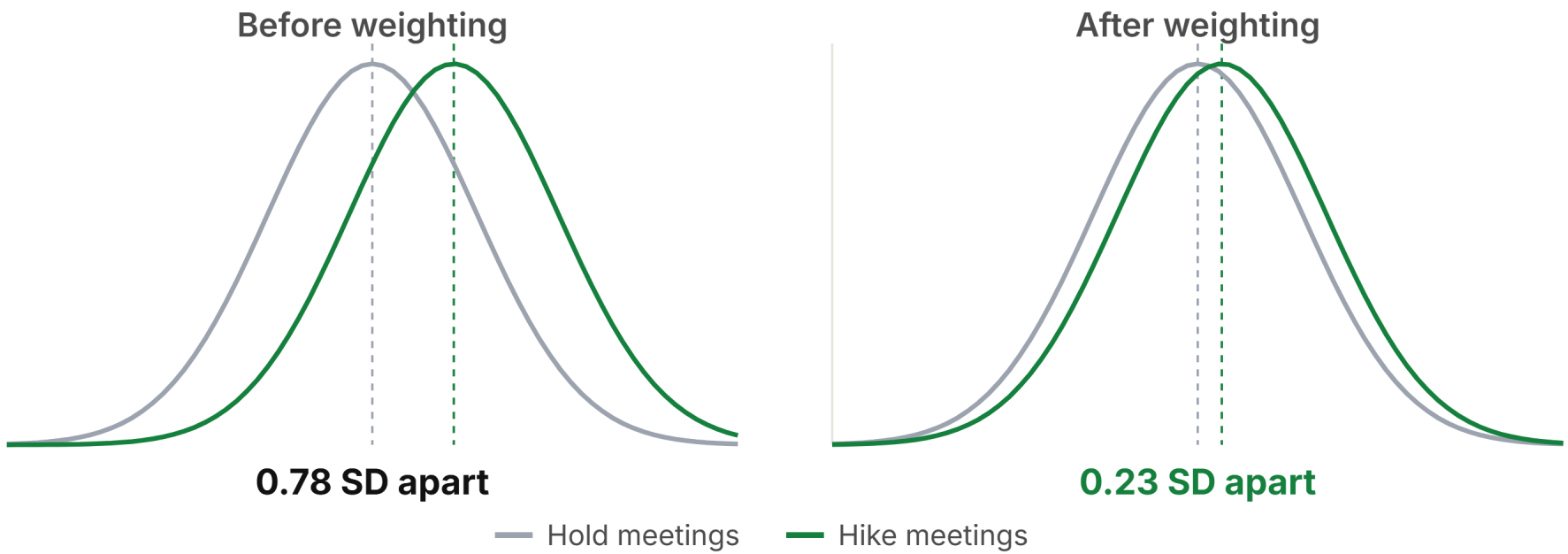
Policy path
recent hikes / cuts

The Fed's mandate
inflation, employment

Market state
VIX, credit spread

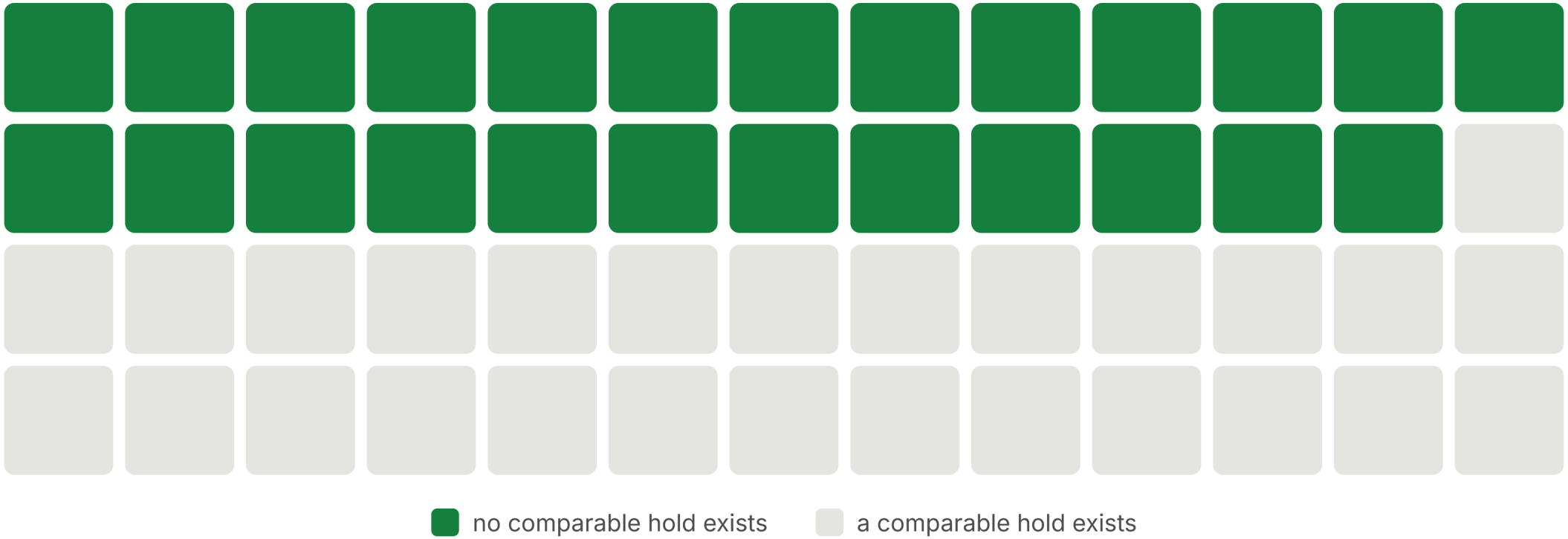
Are the groups comparable?

The confounder gap between hikes and holds, before and after weighting



The central finding

25 of 52 hikes have **no comparable hold** — a near-certain decision, not a close call



How we estimated it

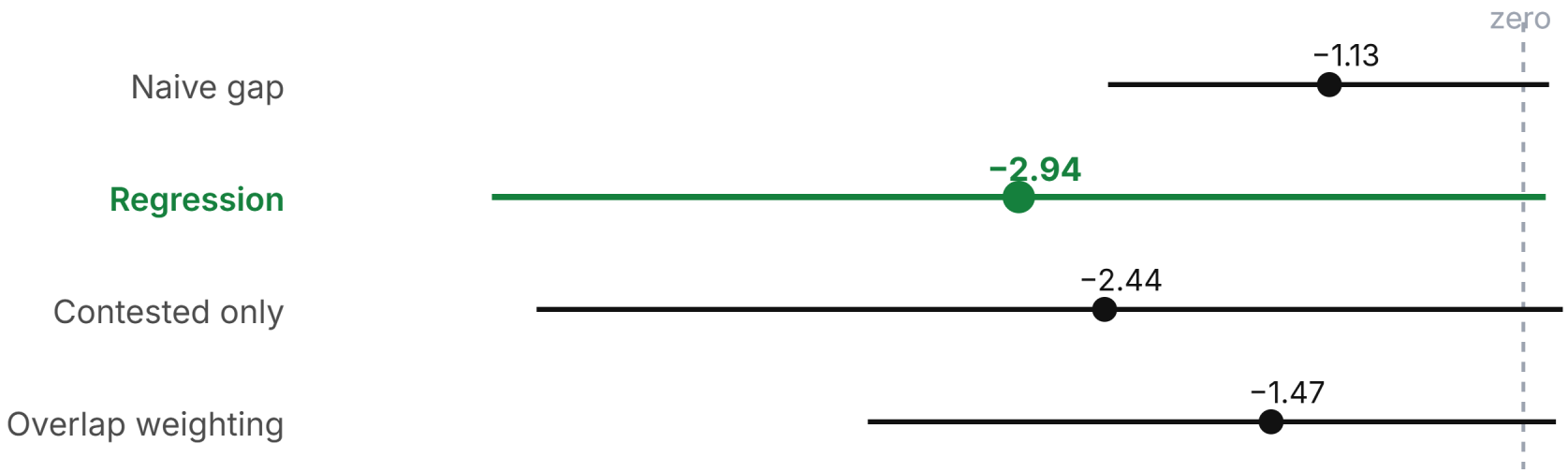
$$Y = \alpha + \beta T + X'\gamma + \varepsilon$$

regression on the 18 confounders, with HC3 robust standard errors

By Frisch–Waugh–Lovell, **β** comes only from the part of the decision the controls did **not** predict — so it leans on the "contested" meetings.

The headline number

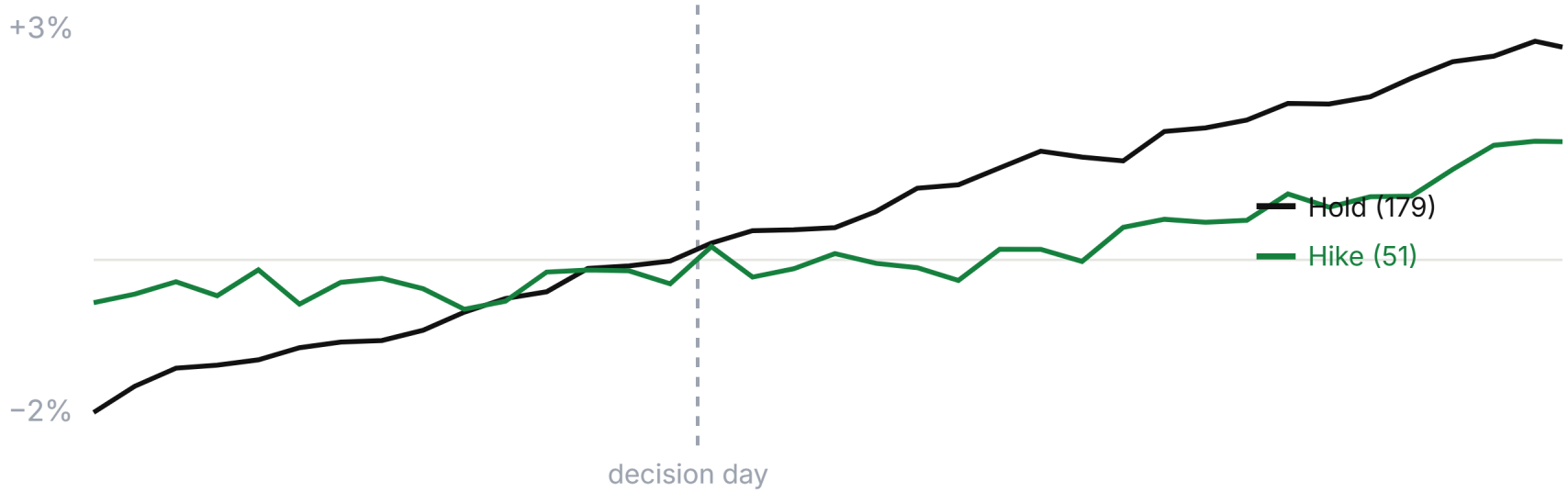
S&P 500 return, 21 days after the decision (97.5% intervals)



Every estimate is negative — and every interval crosses zero.

The effect over time

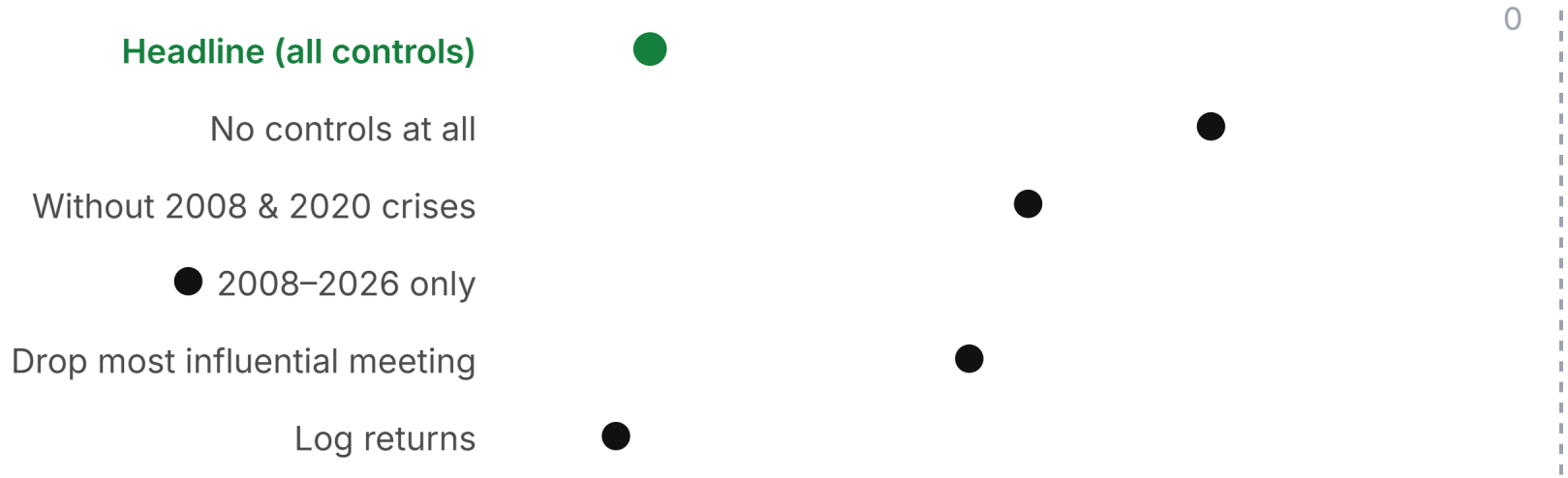
Average S&P 500 path, hike vs. hold meetings



The two paths cross repeatedly — they never clearly separate.

Did it hold up?

21-day estimate with one choice changed at a time



Negative under every choice we tried.

The answer

Condition, fixed in advance	2 days	21 days
1. Regression interval excludes zero	Fail	Fail
2. Contested estimators agree & exclude zero	Fail	Fail
3. Fake-date placebo is clean	Fail	Pass
4. Lead test is clean	Pass	Pass

No causal effect credibly detected — at either horizon.

THE RECOMMENDATION

Don’t trade the announcement.

Treat a lower return as a **stress scenario** to plan for — not a forecast to act on.

The limit

Smallest effect we can detect, at 21 days



4.2 points

Plausible real effect, from the literature



1–3 points

The real effect, if it exists, may be too small for this sample to see — that's a power problem, not evidence of "no effect."

What we’d do next

Size the study to the effect

Pick the horizon and sample from a target effect first — not for business relevance, then discover it's underpowered.

Use a surprise, not a label

Measure how unexpected the decision was, instead of just hike vs. hold — it fixes the overlap problem at the source.

Questions?

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